

Patricia CABEL
Gestionnaire Ecole Doctorale SPIN
UFR S2I
2 rue François Verny
29806 BREST Cedex 9

Review Report Concerning the PhD Dissertation “Event-driven Cycle Navigation for Mobile Robots” submitted by Mr. Quentin Brateau to ENSTA, Brest, France

1 Composition and Structure of the Submitted Thesis

The PhD Thesis “Event-driven Cycle Navigation for Mobile Robots” submitted by Mr. Quentin Brateau, written in English language, is composed of a total of nine chapters.

Chap. 1 gives a brief introduction into the scientific context of challenges in navigation in marine applications. Besides technical challenges, also biological navigation aspects are addressed which are impressive in terms of their simplicity as well as precision and efficiency. These biologically inspired concepts form the basis for parts of the developments performed by Mr. Brateau, especially, when it comes to the stabilization of cyclic navigation patterns.

Thereafter, Chap. 2 summarizes the mathematical and system theoretic background of the work. This chapter comprises an introduction to the mathematical representation of both continuous- and discrete-time dynamic systems, their analysis with respect to controllability and observability, state estimation (on the basis of Luenberger observers), and – lastly – the analysis of stability.

Chap. 3 further introduces the second part of methodological foundations of the submitted work by an introduction into automata theory with a focus on deterministic time automata as well as cyclic automata.

Finally, Chap. 4 concludes the summary of the methodological basics with a recapitulation of set-based methods. Here, the focus is

Fakultät II - Informatik,
Wirtschafts- und Rechtswissenschaften
Department für Informatik

Verteilte Regelung in vernetzten Systemen

Prof. Dr. Andreas Rauh
andreas.rauh@uol.de

Sekretariat
Ingrid Ahlhorn
Tel.: 0441 798 – 2426
Ingrid.ahlhorn@uol.de

Oldenburg, den 21.09.2025



Standort
Industriestraße 11
26121 Oldenburg

Postanschrift
26111 Oldenburg

Paketanschrift
Ammerländer Heerstraße 114–118
26129 Oldenburg

Bankverbindung
Landessparkasse zu Oldenburg
IBAN DE46 2805 0100 0001 9881 12
BIC SLZODE22

Steuernummer
6422008701

www.uol.de

laid on interval approaches, inner and outer approximations of solution sets, and paving strategies with the help of the SIVIA algorithm (set-inversion via interval analysis).

Chap. 5 represents the beginning of the novel contributions developed by Mr. Brateau with a formalization and design of cycle control approaches which are applicable to navigation purposes. The most relevant contributions are the derivations of synchronization conditions as well as a cycle discretization, forming the basis for the introduction of control signals that allow for a stabilization of periodic cycle trajectories. A discussion of different types of controllers (dead-beat approaches, proportional and (regularized) variable-structure controllers) conclude this chapter.

A thorough stability analysis of cycle control approaches on bathymetric maps is then performed in Chap. 6 before the control procedure is extended in Chap. 7 towards a full navigation approach (instead of the stabilization of an individual cyclic trajectory considered before). Most importantly, all derived theoretical approaches are not only tested in numerical simulations but are also validated by means of real-world experiments. These experiments show that the made assumptions also hold in real-life navigation tasks and that the derived cyclic trajectory control is also applicable in outdoor scenarios (experiments performed at the “lac de Guerlédan”) and not only in controlled lab environments.

Chap. 8 introduces set-based state estimation approaches. For these, SIVIA-like algorithms additionally making use of the concepts of contractors and separators are employed. To remove “fake” boundaries between neighboring domains (resulting from different constraints handled at the same time), a union approach of adjacent contractors is derived and tested for its efficiency.

Finally, the PhD Thesis is completed by conclusions and perspectives on possible future research directions in Chap. 9.

2 Thematic Identification and Objective

In contrast to autonomous driving, navigation in marine environments is characterized by a number of additional challenges. These can be characterized, on the one hand, by a much larger influence of external disturbances, e.g. due to currents, waves, and wind. On the other hand, also the access to sensors with a high precision is limited. For example, in under-water applications, the access to GPS-based position and velocity measurement is significantly restricted. Moreover, highly precise maps of the environment on which autonomous vehicles can localize themselves are not always available. Lastly, also absolute velocity measurements are much more complicated to obtain than in road- or rail-based vehicles.

All of these challenges require the implementation of control and navigation algorithms that can cope with disturbances and limited access to sensor data.

Despite this consideration from a technical point of view, recent advances in biological research have shown that animals are capable of navigating with impressive precision over extreme distances. Therefore, nature-inspired approaches for navigation in technical systems gain more and more interest. However, real-life implementations of such techniques are yet not widely available in the context of marine navigation systems.

Therefore, this thesis addresses a highly timely and challenging task of deriving a nature-inspired cyclic navigation approach that can be employed not only to follow trajectories being described by closed loops on the principle of timed automata. In addition, the submitted work aims at robustifying the cycle following by the introduction of a feedback control structure and

by extending it towards a path following task. The latter is represented in the submitted work by following lines of equal depth on a bathymetric map.

Besides the derivation of the control approaches sketched above, the submitted thesis follows the objective to validate all methodologically derived results in real-life applications.

3 Scientific Contribution and Quality of the Presented Results

The scientific contributions achieved by Mr. Brateau are very broad. As discussed in the previous section, these contributions range from the design of a novel cycle control approach, to the experimental validation, and the introduction of a set-based estimation algorithms.

The overall PhD Thesis is very well written, clearly structured and excellently enriched by illustrations and pseudo code that both help to follow complex methodological aspects. It has to be pointed out that in comparison with other PhD Theses, this work contains also a very large portion of experimental validations that significantly add to the high scientific standards of the submitted work.

Moreover, the text of the thesis is written in very good English so that all design decisions can be understood clearly by the interested reader. This thoroughness equally holds for the use of mathematical notations and the derivation of the presented solution approach.

A list of 111 quoted references shows the wide literature review performed by Mr. Brateau and allows for a clear distinction of the new contributions provided by this thesis from the existing state-of-the-art.

Despite my overall judgement of this work as an excellent PhD dissertation, I recommend the correction of the following (minor) mistakes before publishing the final version of the work:

- A very small number of typos needs to be corrected:
 - Abstract: “ta stay stable” should be “to stay stable”
 - Below Fig. 1.1: “is the turtles use of cyclical swimming” should be “is the turtles’ use of cyclical swimming”
 - Remove the repeated word “equation” in Example 2 on p.14 as well as in the text below Fig. 5.1.
 - Please move the punctuation symbols to the last line of multi-line formulas (e.g., (2.27)) and remove indentations if the phrase is not finished after the equation (also after (2.32)).
 - Eq. (2.35) needs to be embedded into the previous phrase.
 - At several places, it is written that “a system is stable on an equilibrium”; This sounds grammatically strange. Either write that the “systems’ trajectories converge in an (asymptotically) stable manner towards the equilibrium” or just say that “the system possesses an asymptotically stable equilibrium”.
 - Please check the punctuation in the last remark on p. 45.
 - Remove the colon before eq. (4.7).
 - Below eq. (5.16): Use the singular form “timed automaton” instead of the plural “timed automata”.
 - The use of singular/plural forms of the verbs needs to be checked below eq. (5.21).
 - Example 29: Typo “evolveg”.
 - Eq. (6.2) needs to be embedded into the previous phrase.
 - Example 34: “This come from” should be “This comes from”.

- P. 24: In the remark, it is said that poles closer to zero lead to faster dynamics. This is only true in discrete-time systems. In the continuous-case, faster dynamics are obtained by moving the poles with negative real parts further away from the imaginary axis. Please either adjust the sentence or include the complete remark (with altered stability domain) in the discussion of discrete-time stability.
- The notation in Def. 24 is misleading: Eq. (2.4) introduces the right-hand side of an ODE as the evolution function of a dynamic systems and its trajectories are denoted as the flow function. However, the thing denoted by f in Def. 24 is also called evolution function, but means the trajectory. Please remove this ambiguity. Such ambiguity exists also in the description of the capture basin.
- Sec. 3.2.2 has the abbreviation DFA been introduced earlier?
- Please introduce the mathematical definition of “width” in Algorithm 3.
- Algorithm 4: Shouldn’t x in line 12 be x_{out} ? This also appears in line 14 of Algorithm 5.
- In the second line of Sec. 6.2, a reference to Chap. 7 is given. I think this should be Chap. 5.
- Sec. 6.2 and Fig. 6.3: The “transport of measurements” needs to be better explained. Moreover, give guidelines on how to choose measurements (their numbers and their required features).
- For Chap. 8, I recommend to add a short note that better explains the relation of the proposed estimation scheme to the cycle-based navigation.

4 Concluding Assessment

In conclusion, it can be stated that the submitted Thesis of Mr. Quentin Brateau is of very high scientific quality. Especially, the thorough derivation and experimental validation of a novel cycle-based control and navigation scheme should be acknowledged. I deeply hope that besides this PhD dissertation, further publications will originate from this work that demonstrate the outcomes of this research to the scientific community.

In view of all aspects above, I recommend to accept this Doctoral Thesis without any doubt. Moreover, I recommend to proceed with the oral defense of the work.

Before a final publication of this Thesis, I kindly ask Mr. Brateau to carefully consider the suggestions for minor corrections summarized in Sec. 3 of this review.

Oldenburg, 21st of September, 2025



Prof. Dr.-Ing. habil. Andreas Rauh
(head of the research group
Distributed Control in Interconnected Systems,
Carl von Ossietzky Universität Oldenburg)